



# MRK CONSTRUCTION

## DESIGN & BUILD

Subject: MEPP COST PROPOSAL  
Project: UBIQUITY AYALA MALL BACOLOD OFFICE  
Location: 4F Ayala Malls Capital Central Gatuslao Street  
Bacolod City

| DIVISION CODE | DESCRIPTION                                                                                               | Trade Responsible | Unit | Whole System |            |              |
|---------------|-----------------------------------------------------------------------------------------------------------|-------------------|------|--------------|------------|--------------|
|               |                                                                                                           |                   |      | Quantity     | Unit Cost  | Amount       |
| 15.00         | <b>Electrical Trade Package</b>                                                                           |                   |      |              |            |              |
|               | <b>Basic Materials and Methods</b>                                                                        |                   |      |              |            |              |
|               | <b>Panelboards, Transformers, UPS and Disconnect Switches</b>                                             |                   |      |              |            |              |
|               | To supply and install panelboards, transformers and disconnect switches, as per drawing and specification |                   |      |              |            |              |
|               | MTS-1200A                                                                                                 | Elec              | set  | 2            | 260,000.00 | 520,000.00   |
|               | ECB-1200A (with SMART kWhr meter, BMS Ready)                                                              | Elec              | set  | 2            | 175,000.00 | 350,000.00   |
|               | MDP-4F                                                                                                    | Elec              | set  | 2            | 370,000.00 | 740,000.00   |
|               | DP-4F-01 (with SMART kWhr meter, BMS Ready)                                                               | Elec              | set  | 2            | 145,000.00 | 290,000.00   |
|               | LP-4F-1A                                                                                                  | Elec              | set  | 2            | 73,000.00  | 146,000.00   |
|               | LP-4F-1B                                                                                                  | Elec              | set  | 2            | 73,000.00  | 146,000.00   |
|               | PP-4F-1A                                                                                                  | Elec              | set  | 2            | 100,000.00 | 200,000.00   |
|               | PP-4F-1B                                                                                                  | Elec              | set  | 2            | 100,000.00 | 200,000.00   |
|               | PP-4F-WS-1A                                                                                               | Elec              | set  | 2            | 73,000.00  | 146,000.00   |
|               | PP-4F-WS-1B                                                                                               | Elec              | set  | 2            | 73,000.00  | 146,000.00   |
|               | UPP-4F-A                                                                                                  | Elec              | set  | 2            | 24,500.00  | 49,000.00    |
|               | UPP-4F-B                                                                                                  | Elec              | set  | 2            | 24,500.00  | 49,000.00    |
|               | PP-4F-AC-01                                                                                               | Elec              | set  | 2            | 120,000.00 | 240,000.00   |
|               | PP-ACCU-4F-01 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | PP-ACCU-4F-02 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | PP-ACCU-4F-03 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | PP-ACCU-4F-04 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | PP-ACCU-4F-05 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | PP-ACCU-4F-06 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | PP-ACCU-4F-07 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-ACCU-4F-HUB-01 in NEMA 3R                                                                             | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-ACCU-4F-HUB-02 in NEMA 3R                                                                             | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-ACCU-4F-AVR-01 in NEMA 3R                                                                             | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-ACCU-4F-AVR-02 in NEMA 3R                                                                             | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-HUB-4F-RACK 1 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-HUB-4F-RACK 2 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-HUB-4F-RACK 3 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-HUB-4F-RACK 4 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-KIOSK-1A in NEMA 1                                                                                    | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-KIOSK-1B in NEMA 1                                                                                    | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-PFCU-4F-06-01 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-PFCU-4F-06-02 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-PFCU-4F-06-03 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-PFCU-4F-07-01 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-PFCU-4F-07-02 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-PFCU-4F-07-03 in NEMA 1                                                                               | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | ECB-SEF-5F-01 in NEMA 3R                                                                                  | Elec              | set  | 2            | 11,000.00  | 22,000.00    |
|               | 500kVA 400/230 V, 3 phase, 60Hz Automatic Voltage Regulator                                               | Elec              | set  |              | -          | -            |
|               | 10kVA 230 V, Single Phase, 60Hz, Rack-mounted UPS (10 Modules)                                            | Elec              | set  | 2            | 500,000.00 | 1,000,000.00 |
|               | Disconnect Switch, 20AT, 1P, 230V in Nema-1                                                               | Elec              | set  | 180          | 6,100.00   | 1,098,000.00 |
|               | Digital kW-hr meter, BMS Ready                                                                            | Elec              | set  | 2            | 60,000.00  | 120,000.00   |
|               | TVSS                                                                                                      | Elec              | set  |              |            |              |
|               |                                                                                                           |                   |      |              | -          |              |
|               | Others (Required to complete the system kindly specify below)                                             |                   |      |              | -          |              |
|               | 1200AT Plug In CB, 400V, 65KAIC                                                                           | Elec              | set  | 4            | 800,000.00 | 3,200,000.00 |
|               | LTFC - (for 4 x 200mm2)                                                                                   | Elec              |      | 40           | 255.00     | 10,200.00    |

|  |                                                                                                                                                                                             |      |     |       |            |                      |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|-------|------------|----------------------|
|  | ECB-AVR-4F-PH1-PRI<br>600AT, 3P, 400V, 60Hz<br>IN NEMA-1                                                                                                                                    | Elec | set | 4     | 80,000.00  | 320,000.00           |
|  | AVR-4F-PH1<br>400kVA, 3P,<br>400V, 60Hz, AVR                                                                                                                                                | Elec | set | 2     | 510,000.00 | 1,020,000.00         |
|  | Consumables                                                                                                                                                                                 | Elec | set | 2     | 143,000.00 | 286,000.00           |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  | <b>Sub-Total Section : Panelboards, Transformer and Disconnect Switches</b>                                                                                                                 |      |     |       | -          | <b>10,804,200.00</b> |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  | <b>Wires &amp; Cables and Wiring Devices</b>                                                                                                                                                |      |     |       | -          |                      |
|  | To supply and install the following copper cables with insulated (green/yellow colour) circuit protective conductors in trunking and conduit (trunking, conduit to be measured separately). |      |     |       | -          |                      |
|  | THHN Cables                                                                                                                                                                                 |      |     |       | -          |                      |
|  | 3.5mm <sup>2</sup>                                                                                                                                                                          | Elec | lm  | 44466 | 20.00      | 889,320.00           |
|  | 5.5mm <sup>2</sup>                                                                                                                                                                          | Elec | lm  | 4080  | 30.00      | 122,400.00           |
|  | 8.0mm <sup>2</sup>                                                                                                                                                                          | Elec | lm  | 6926  | 45.00      | 311,670.00           |
|  | 14mm <sup>2</sup>                                                                                                                                                                           | Elec | lm  | 1888  | 73.00      | 137,824.00           |
|  | 22mm <sup>2</sup>                                                                                                                                                                           | Elec | lm  | 6564  | 114.00     | 748,296.00           |
|  | 30mm <sup>2</sup>                                                                                                                                                                           | Elec | lm  | 4080  | 165.00     | 673,200.00           |
|  | 200mm <sup>2</sup>                                                                                                                                                                          | Elec | lm  | 3376  | 980.00     | 3,308,480.00         |
|  | 250mm <sup>2</sup>                                                                                                                                                                          | Elec | lm  | 144   | 1,260.00   | 181,440.00           |
|  |                                                                                                                                                                                             |      |     | 0     | -          | -                    |
|  | Others (Required to complete the system kindly specify below)                                                                                                                               | Elec | lm  |       | -          | -                    |
|  | 38mm <sup>2</sup>                                                                                                                                                                           | Elec | lm  |       |            | -                    |
|  | 80mm <sup>2</sup>                                                                                                                                                                           | Elec | lm  |       |            | -                    |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  | <b>Sub-Total Section : Wires and Cables</b>                                                                                                                                                 |      |     |       | -          | <b>6,372,630.00</b>  |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  | To supply and install the following wiring devices                                                                                                                                          |      |     |       | -          |                      |
|  | 1 Gang Toggle Switch 1P 15A, 250V                                                                                                                                                           | Elec | set | 100   | 121.00     | 12,100.00            |
|  | 2 Gang Toggle Switch, 1P, 15A, 250V                                                                                                                                                         | Elec | set | 102   | 185.00     | 18,870.00            |
|  | 3 Gang Toggle Switch, 1P, 15A, 250V                                                                                                                                                         | Elec | set | 8     | 250.00     | 2,000.00             |
|  | Duplex Outlet 15A, 250V 2P, 3W Universal Grounding Type                                                                                                                                     | Elec | set |       |            | -                    |
|  | Duplex Outlet 15A, 250V 2P, 3W Universal Grounding Type - GFCI                                                                                                                              | Elec | set | 56    | 3,575.00   | 200,200.00           |
|  | Table Mounted, Duplex Outlet 15A, 250V 2P, 3W Universal Grounding Type                                                                                                                      | Elec | set | 1058  | 305.00     | 322,690.00           |
|  | Floor Mounted, Duplex Outlet 15A, 250V 2P, 3W Universal Grounding Type                                                                                                                      | Elec | set | 82    | 1,865.00   | 152,930.00           |
|  | Single Outlet, Twistlock 32A, 250V, 2P, 3W Grounding Type (Mennekes IEC 309) - Verify w/ IT                                                                                                 | Elec | set | 42    | 2,844.00   | 119,448.00           |
|  | Special Purpose Outlet, SPN CEE FORM, 20A, 250V, 2P,3W, Grounding Type                                                                                                                      | Elec | set | 60    | 6,905.00   | 414,300.00           |
|  |                                                                                                                                                                                             |      |     |       | -          | -                    |
|  | Others (Required to complete the system kindly specify below)                                                                                                                               | Elec | set |       | -          | -                    |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  | <b>Sub-Total Section : Wiring Devices</b>                                                                                                                                                   |      |     |       | -          | <b>1,242,538.00</b>  |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  |                                                                                                                                                                                             |      |     |       | -          |                      |
|  | <b>Lighting Fixture</b>                                                                                                                                                                     |      |     |       | -          |                      |
|  | To supply and install the following lighting fixture with LED Driver                                                                                                                        |      |     |       | -          |                      |
|  | 18 Watts LED Downlight (Refer to I.D. for Specifications)                                                                                                                                   | Elec | set | 1988  | 1,460.00   | 2,902,480.00         |
|  | Decorative Light 1 (Refer to I.D. for Specifications)                                                                                                                                       | Elec | set | 84    | 2,100.00   | 176,400.00           |
|  | Decorative Light 2 (Refer to I.D. for Specifications)                                                                                                                                       | Elec | set | 72    | 2,100.00   | 151,200.00           |
|  | 600mm X 600mm 28 Watts LED Panel Light                                                                                                                                                      | Elec | set | 22    | 2,100.00   | 46,200.00            |
|  | 600mm X 1200mm 42 Watts LED Panel Light                                                                                                                                                     | Elec | set | 904   | 2,100.00   | 1,898,400.00         |
|  | 25 Watts / Meter LED Linear Lamp (Refer to I.D. for Specifications)                                                                                                                         | Elec | set | 128   | 723.00     | 92,544.00            |

|  |                                                                                                                                         |      |      |       |            |                     |
|--|-----------------------------------------------------------------------------------------------------------------------------------------|------|------|-------|------------|---------------------|
|  | LED Strip (Refer to I.D. for Specifications)                                                                                            | Elec | set  | 158   | 3,575.00   | 564,850.00          |
|  | Illuminated Maintained Emergency Light Exit Sign w/ 2 Hr capability                                                                     | Elec | set  | 36    | 2,100.00   | 75,600.00           |
|  | Battery Pack                                                                                                                            | Elec | set  | 320   | 5,030.00   | 1,609,600.00        |
|  | Others (Required to complete the system kindly specify below)                                                                           | Elec | set  | 2     | 203,000.00 | 406,000.00          |
|  | Miscellaneous, Consumables and Supports                                                                                                 |      |      |       | -          | -                   |
|  |                                                                                                                                         |      |      |       | -          | -                   |
|  |                                                                                                                                         |      |      |       | -          | -                   |
|  | <b>Sub-Total Section : Lighting Fixture</b>                                                                                             |      |      |       | -          | <b>7,923,274.00</b> |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  | <b>Grounding System</b>                                                                                                                 |      |      |       | -          |                     |
|  | To supply and install the grounding system                                                                                              |      |      |       | -          |                     |
|  | Copper Ground Bus                                                                                                                       | Elec | lm   |       | -          | -                   |
|  | Grounding Clip                                                                                                                          | Elec | lm   |       | -          | -                   |
|  | 3.5mm <sup>2</sup> THHN                                                                                                                 | Elec | lm   | 36546 | 20.00      | 730,920.00          |
|  | 5.5mm <sup>2</sup> THHN                                                                                                                 | Elec | lm   | 1250  | 30.00      | 37,500.00           |
|  | 8.0mm <sup>2</sup> THHN                                                                                                                 | Elec | lm   | 2100  | 45.00      | 94,500.00           |
|  | 50mm <sup>2</sup> THHN                                                                                                                  | Elec | lm   |       |            | -                   |
|  | 80mm <sup>2</sup> THHN                                                                                                                  | Elec | lm   | 128   | 415.00     | 53,120.00           |
|  | Others (Required to complete the system kindly specify below)                                                                           | Elec | lm   |       | -          | -                   |
|  | 14mm <sup>2</sup> THHN                                                                                                                  | Elec | lm   |       |            | -                   |
|  | 22mm <sup>2</sup> THHN                                                                                                                  | Elec | lm   |       |            | -                   |
|  | 30mm <sup>2</sup> THHN                                                                                                                  | Elec | lm   | 36    | 166.00     | 5,976.00            |
|  | 38mm <sup>2</sup> THHN                                                                                                                  | Elec | lm   | 452   | 242.00     | 109,384.00          |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  | <b>Sub-Total Section Grounding System</b>                                                                                               |      |      |       | -          | <b>1,031,400.00</b> |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  | <b>Raceways and Wireways (Small power and Lighting Systems)</b>                                                                         |      |      |       | -          |                     |
|  | To supply and install cable trunking 1.6mm minimum electrogalvanised sheet steel finished in oven-baked electrostatically coated epoxy. |      |      |       | -          |                     |
|  | 100mm (W) x 50mm (D)                                                                                                                    | Elec | lm   | 40    | 780.00     | 31,200.00           |
|  | 200mm (W) x 100mm (D)                                                                                                                   | Elec | lm   | 50    | 1,325.00   | 66,250.00           |
|  | 300mm (W) x 100mm (D)                                                                                                                   | Elec | lm   | 240   | 1,684.00   | 404,160.00          |
|  | 400mm (W) x 100mm (D)                                                                                                                   | Elec | lm   | 160   | 1,045.00   | 167,200.00          |
|  | 500mm (W) x 100mm (D)                                                                                                                   | Elec | lm   | 24    | 2,400.00   | 57,600.00           |
|  |                                                                                                                                         |      |      |       | -          | -                   |
|  | Others (Required to complete the system kindly specify below)                                                                           |      |      |       | -          | -                   |
|  | 500mm (W) x 100mm (D) - H Elbow                                                                                                         | Elec | ea   | 2     | 2,800.00   | 5,600.00            |
|  | 400mm (W) x 100mm (D) - V Elbow                                                                                                         | Elec | ea   | 4     | 2,900.00   | 11,600.00           |
|  | 400mm (W) x 100mm (D) - H Elbow                                                                                                         | Elec | ea   | 8     | 2,330.00   | 18,640.00           |
|  | 300mm (W) x 100mm (D) - H Elbow                                                                                                         | Elec | ea   | 16    | 1,845.00   | 29,520.00           |
|  | 100mm (W) x 50mm (D) - H Elbow                                                                                                          | Elec | ea   | 6     | 923.00     | 5,538.00            |
|  |                                                                                                                                         |      |      |       |            |                     |
|  |                                                                                                                                         |      |      |       |            |                     |
|  | To supply and install the following conduits complete with saddles, clips, tees, elbows long bends, cross boxes, etc.                   |      |      |       | -          | -                   |
|  | IMC Conduits                                                                                                                            |      |      |       | -          | -                   |
|  | a.1 15mm diameter                                                                                                                       | Elec | lm   | 34460 | 122.00     | 4,204,120.00        |
|  | a.2 20mm diameter                                                                                                                       | Elec | lm   | 3468  | 122.00     | 423,096.00          |
|  | a.3 25mm diameter                                                                                                                       | Elec | lm   | 3456  | 190.00     | 656,640.00          |
|  | a.4 32mm diameter                                                                                                                       | Elec | lm   | 104   | 298.00     | 30,992.00           |
|  | a.5 80mm diameter                                                                                                                       | Elec | lm   | 144   | 867.00     | 124,848.00          |
|  |                                                                                                                                         |      |      | 0     | -          | -                   |
|  | Others (Required to complete the system kindly specify below)                                                                           | Elec | lm   | 0     | -          | -                   |
|  | 20mm diameter                                                                                                                           | Elec | lm   | 24    | 122.00     | 2,928.00            |
|  | Threaded Rod, Expansion Bolts, Unistruts, consumables                                                                                   | Elec | sets | 514   | 305.00     | 156,770.00          |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  | <b>Sub-Total Section : Raceway and wireways (Small power and Lighting)</b>                                                              |      |      |       | -          | <b>6,396,702.00</b> |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  |                                                                                                                                         |      |      |       | -          |                     |
|  | <b>Other Items Required (Please Specify)</b>                                                                                            |      |      |       | -          |                     |
|  | Upgrade and Revision to existing BUS WAY not included.                                                                                  |      |      |       | -          |                     |

|              |                                                                                                                                                     |             |     |              |            |                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----|--------------|------------|----------------------|
|              |                                                                                                                                                     |             |     |              | -          |                      |
|              | <b>Grand Total Electrical System Package</b>                                                                                                        |             |     |              |            | <b>33,770,744.00</b> |
|              |                                                                                                                                                     |             |     |              |            |                      |
| <b>16.00</b> | <b>Electronics Trade Package</b>                                                                                                                    |             |     | <b>Total</b> |            |                      |
|              | <b>Electronics Works</b>                                                                                                                            |             |     |              |            |                      |
|              | <b>Basic Materials and Methods</b>                                                                                                                  |             |     |              |            |                      |
|              | <b>Fire Detection and Alarm System</b>                                                                                                              |             |     |              |            |                      |
|              | To supply and install the complete fire detection and alarm system and interface with base building FDAS (To match existing Basebuilt FDAS System). |             |     |              |            |                      |
|              | Fire Alarm Control Panel (c/w power supply & back-up battery)                                                                                       | Electronics | lot | 2            | 125,000.00 | 250,000.00           |
|              | EVAC Panel (c/w power supply & back-up battery)                                                                                                     | Electronics | lot | 2            | 125,000.00 | 250,000.00           |
|              | Interface Module                                                                                                                                    | Electronics | set | 2            | 125,000.00 | 250,000.00           |
|              | Addressable Photoelectric Smoke Detector                                                                                                            | Electronics | set | 240          | 4,000.00   | 960,000.00           |
|              | Addressable Photoelectric Smoke Detector with Sounder Base                                                                                          | Electronics | set |              | -          | -                    |
|              | Addressable Heat Detector (Combination Rate of Rise & Fixed Temperature)                                                                            | Electronics | set |              | -          | -                    |
|              | Addressable Input Module                                                                                                                            | Electronics | set | 18           | 12,000.00  | 216,000.00           |
|              | Addressable Output Module                                                                                                                           | Electronics | set | 8            | 12,000.00  | 96,000.00            |
|              | Addressable Manual Pull Station                                                                                                                     | Electronics | set |              | -          | -                    |
|              | Fire Alarm Speaker with Strobe Light                                                                                                                | Electronics | set |              | -          | -                    |
|              | Fireman's Telephone Jack                                                                                                                            | Electronics | set |              | -          | -                    |
|              | Synchronization Panel                                                                                                                               | Electronics | set | 2            | -          | -                    |
|              | #18 2/C Fire Alarm Cable Shielded Twisted Pair Wires (Fire Rated)                                                                                   | Electronics | set | 1500         | 140.00     | 210,000.00           |
|              | #16 2/C Fire Alarm Cable Shielded Twisted Pair Wires (Fire Rated)                                                                                   | Electronics | set | 1500         | 135.00     | 202,500.00           |
|              | 25mm dia. IMC                                                                                                                                       | Electronics | lm  | 1500         | 236.00     | 354,000.00           |
|              | 50mm dia. IMC                                                                                                                                       | Electronics | lm  |              | 1,180.00   | -                    |
|              | Other Accessories for the complete operation of the system                                                                                          | Electronics |     |              | -          | -                    |
|              | Interface to building FDAS                                                                                                                          | Electronics | lot | 2            | 45,000.00  | 90,000.00            |
|              | Interface to building EVAC System                                                                                                                   | Electronics | lot | 2            | 45,000.00  | 90,000.00            |
|              | Interface to BMS                                                                                                                                    | Electronics | lot | 2            | 45,000.00  | 90,000.00            |
|              | Interface Cables to other life safety equipments                                                                                                    | Electronics | set | 2            | 45,000.00  | 90,000.00            |
|              | Others (Please specify in details)                                                                                                                  | Electronics | set |              | -          | -                    |
|              | As-built drawings                                                                                                                                   | Electronics | lot | 2            | 35,500.00  | 71,000.00            |
|              | Operation and Maintenance Manual                                                                                                                    | Electronics | lot | 2            | 35,500.00  | 71,000.00            |
|              | Testing and Commissioning                                                                                                                           | Electronics | lot | 2            | 50,000.00  | 100,000.00           |
|              |                                                                                                                                                     |             |     |              | -          |                      |
|              | <b>Sub-Total Section : Fire Detection and Alarm System</b>                                                                                          |             |     |              | -          | <b>3,390,500.00</b>  |
|              |                                                                                                                                                     |             |     |              | -          |                      |
|              | <b>Background Music/Public Address System</b>                                                                                                       |             |     |              | -          |                      |
|              | To supply and install the complete Public Address and Background Music System:                                                                      |             |     |              | -          |                      |
|              | 6W Speaker, Ceiling Mounted (multi tap, c/w fire rated)                                                                                             | Electronics | ea  | 280          | 2,200.00   | 616,000.00           |
|              | 6W Speaker, Wall Mounted (multi tap)                                                                                                                | Electronics | ea  |              |            | -                    |
|              | #18 2/C PA Cable Shielded Twisted Pair Wires (Fire Rated)                                                                                           | Electronics | set | 1500         | 130.00     | 195,000.00           |
|              | #16 2/C PA Alarm Cable Shielded Twisted Pair Wires (Fire Rated)                                                                                     | Electronics | set | 1500         | 125.00     | 187,500.00           |
|              |                                                                                                                                                     |             |     |              | -          | -                    |
|              | Equipments                                                                                                                                          |             |     |              | -          | -                    |
|              | PA Rack                                                                                                                                             | Electronics | set | 2            | 70,000.00  | 140,000.00           |
|              | Media Player w/ tuner                                                                                                                               | Electronics | set | 2            | 37,000.00  | 74,000.00            |
|              | Paging Microphone Equipment                                                                                                                         | Electronics | set | 2            | 27,800.00  | 55,600.00            |
|              | Pre-Amplifier                                                                                                                                       | Electronics | set | 2            | 80,000.00  | 160,000.00           |
|              | Power Amplifier                                                                                                                                     | Electronics | set | 2            | 41,000.00  | 82,000.00            |
|              | UPS (1Kva,230V)                                                                                                                                     | Electronics | set | 2            | 13,000.00  | 26,000.00            |
|              | Interfacing Modules/Cables                                                                                                                          | Electronics | set | 2            | 18,500.00  | 37,000.00            |
|              | DSP Matrix Controller                                                                                                                               | Electronics | set | 2            | 70,000.00  | 140,000.00           |
|              |                                                                                                                                                     |             |     |              | -          | -                    |
|              | Roughing-ins (All necessary fittings, supports and other accessories are already included from straight pipes)                                      |             |     |              | -          | -                    |
|              | 25mmØ IMC                                                                                                                                           | Electronics | lm  | 1500         | 130.00     | 195,000.00           |
|              | 50mmØ IMC                                                                                                                                           | Electronics | lm  |              | 125.00     | -                    |
|              | Pullboxes & Wireways                                                                                                                                | Electronics | lot | 1500         | 325.00     | 487,500.00           |
|              | Other Accessories for the complete operation of the system                                                                                          | Electronics | set |              | -          | -                    |
|              | Interface to building EVAC System                                                                                                                   | Electronics | lot | 2            | 125,000.00 | 250,000.00           |
|              | Others (Please specify in details)                                                                                                                  | Electronics | set |              | -          | -                    |
|              | As-built drawings                                                                                                                                   | Electronics | lot | 2            | 27,800.00  | 55,600.00            |
|              | Operation and Maintenance Manual                                                                                                                    | Electronics | lot | 2            | 18,500.00  | 37,000.00            |

|  |                                                                                                                                            |             |      |       |            |                     |
|--|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|-------|------------|---------------------|
|  | Testing and Commissioning                                                                                                                  | Electronics | lot  | 2     | 125,000.00 | 250,000.00          |
|  |                                                                                                                                            |             |      |       | -          |                     |
|  | <b>Sub-Total Section : Background Music/Public Address System</b>                                                                          |             |      |       | -          | <b>2,988,200.00</b> |
|  |                                                                                                                                            |             |      |       | -          |                     |
|  |                                                                                                                                            |             |      |       | -          |                     |
|  | <b>Raceways and Wireways (Telecommunication and Structured Cabling-Roughing-ins only)</b>                                                  |             |      |       | -          |                     |
|  | To supply and install cable trunking 1.6mm minimum electrogalvanised sheet steel finished in oven-baked electrostatically coated epoxy.    |             |      |       | -          |                     |
|  | 75mm (W) x 100mm (D)                                                                                                                       | Electronics | lm   |       |            | -                   |
|  | 100mm (W) x 100mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | 125mm (W) x 100mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | 175mm (W) x 100mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | 75mm (W) x 150mm (D)                                                                                                                       | Electronics | lm   |       |            | -                   |
|  | 100mm (W) x 150mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | 125mm (W) x 150mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | 150mm (W) x 150mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | 350mm (W) x 150mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | 75mm (W) x 250mm (D)                                                                                                                       | Electronics | lm   |       |            | -                   |
|  | Others (Please specify in details)                                                                                                         | Electronics | lm   |       | -          | -                   |
|  | 200mm (W) x 150mm (D)                                                                                                                      | Electronics | lm   | 260   | 1,350.00   | 351,000.00          |
|  | 200mm (W) x 150mm (D) - H Elbow                                                                                                            | Electronics | lm   | 12    | 1,900.00   | 22,800.00           |
|  | 200mm (W) x 150mm (D) - V Elbow                                                                                                            | Electronics | lm   | 4     | 1,900.00   | 7,600.00            |
|  |                                                                                                                                            |             |      |       | -          | -                   |
|  | 100mm (W) x 50mm (D)                                                                                                                       | Electronics | lm   | 180   | 800.00     | 144,000.00          |
|  | 150mm (W) x 50mm (D)                                                                                                                       | Electronics | lm   | 40    | 1,380.00   | 55,200.00           |
|  | 200mm (W) x 50mm (D)                                                                                                                       | Electronics | lm   | 50    | 1,380.00   | 69,000.00           |
|  | 250mm (W) x 50mm (D)                                                                                                                       | Electronics | lm   | 90    | 1,750.00   | 157,500.00          |
|  | 300mm (W) x 50mm (D)                                                                                                                       | Electronics | lm   | 140   | 1,750.00   | 245,000.00          |
|  | 200mm (W) x 100mm (D)                                                                                                                      | Electronics | lm   | 230   | 1,750.00   | 402,500.00          |
|  |                                                                                                                                            |             |      |       | -          | -                   |
|  | To supply and install the following conduits complete with saddles, clips, tees, elbows long bends, cross boxes, etc.                      |             |      |       | -          | -                   |
|  | IMC Conduits                                                                                                                               |             |      |       | -          | -                   |
|  | a.1 15mm diameter                                                                                                                          | Electronics | lm   |       |            | -                   |
|  | a.2 20mm diameter                                                                                                                          | Electronics | lm   |       |            | -                   |
|  | a.3 25mm diameter                                                                                                                          | Electronics | lm   |       |            | -                   |
|  | a.3 25mm diameter                                                                                                                          | Electronics | lm   |       |            | -                   |
|  | a.3 25mm diameter                                                                                                                          | Electronics | lm   |       |            | -                   |
|  | a.4 32mm diameter                                                                                                                          | Electronics | lm   |       |            | -                   |
|  | a.5 50mm diameter                                                                                                                          | Electronics | lm   | 760   | 385.00     | 292,600.00          |
|  | Junction Box                                                                                                                               | Electronics | lm   |       |            | -                   |
|  | Others (Please specify in details)                                                                                                         | Electronics | lm   |       |            | -                   |
|  | 300mm (W) x 100mm (D)                                                                                                                      | Electronics | lm   |       |            | -                   |
|  | Threaded Rod, Expansion Bolts, Unistruts, consumables                                                                                      | Electronics | sets | 1750  | 315.00     | 551,250.00          |
|  |                                                                                                                                            |             |      |       | -          | -                   |
|  | <b>WIRES AND ACCESSORIES</b>                                                                                                               |             |      |       | -          | -                   |
|  | 8 core fiber optic cable (OM 4 Multimode)                                                                                                  |             |      |       |            |                     |
|  | Cat6 patch cord (Nexans)                                                                                                                   |             |      |       |            |                     |
|  | Fiber patch cord (Nexans)                                                                                                                  |             |      |       |            |                     |
|  | CAT6 UTP Cable                                                                                                                             |             | lm   | 40700 | 25.00      | 1,017,500.00        |
|  | Single Port Universal IO                                                                                                                   |             | sets | 880   | 325.00     | 286,000.00          |
|  | Dual Port Universal IO                                                                                                                     |             | sets | 42    | 338.00     | 14,196.00           |
|  | CAT6 UNSHIELDED KEYSTONE JACK (90DEGREE)                                                                                                   |             | sets | 1928  | 425.00     | 819,400.00          |
|  | Utility Box                                                                                                                                |             | sets | 922   | 210.00     | 193,620.00          |
|  |                                                                                                                                            |             |      |       | -          | -                   |
|  |                                                                                                                                            |             |      |       | -          | -                   |
|  |                                                                                                                                            |             |      |       | -          | -                   |
|  |                                                                                                                                            |             |      |       | -          | -                   |
|  | <b>Sub-Total Section : Raceways and Wireways (Telecommunication and Structured Cabling-Roughing-ins only)</b>                              |             |      |       | -          | <b>4,629,166.00</b> |
|  |                                                                                                                                            |             |      |       | -          |                     |
|  | <b>Access Control and Monitoring System (Rough-ins Only)</b>                                                                               |             |      |       | -          |                     |
|  | To supply and install the complete Access Control & Intrusion Alarm System including Security software, programming and FDAS interfacing.: |             |      |       | -          |                     |
|  | Security Workstation                                                                                                                       | Electronics | set  | 2     | 95,000.00  | 190,000.00          |
|  | Access Control Panel                                                                                                                       | Electronics | set  | 2     | 110,000.00 | 220,000.00          |
|  | Power Supply with UPS                                                                                                                      | Electronics | set  | 2     | 11,100.00  | 22,200.00           |

|  |                                                                                     |             |     |      |            |                     |
|--|-------------------------------------------------------------------------------------|-------------|-----|------|------------|---------------------|
|  | Switch                                                                              | Electronics | set | 2    | 30,208.75  | 60,417.50           |
|  | Door Contact                                                                        | Electronics | set | 14   | 22,500.00  | 315,000.00          |
|  | Electromagnetic Lock                                                                | Electronics | set | 14   | 2,300.00   | 32,200.00           |
|  | Card Reader                                                                         | Electronics | set | 14   | 2,300.00   | 32,200.00           |
|  | Emergency Break Glass                                                               | Electronics | set | 14   | 1,400.00   | 19,600.00           |
|  | Request to Exist Button                                                             | Electronics | set | 14   | 695.00     | 9,730.00            |
|  | Audio-Visual Notification Appliance                                                 | Electronics | set | 2    | 23,000.00  | 46,000.00           |
|  | Bypass Key Switch                                                                   | Electronics | set | 14   | 2,300.00   | 32,200.00           |
|  | Duress Alarm Button                                                                 | Electronics | set | 14   | 2,300.00   | 32,200.00           |
|  | #18 2/C Security Communication Cable Shielded Twisted Pair Wires                    | Electronics | set | 1050 | 130.00     | 136,500.00          |
|  | #16 2/C Security Communication Cable Shielded Twisted Pair Wires                    | Electronics | set |      | -          | -                   |
|  | Others (Please specify in details)                                                  | Electronics | set |      | -          | -                   |
|  |                                                                                     |             |     |      | -          | -                   |
|  | Supply and Installation of Conduit Fittings and Supports                            |             |     |      | -          | -                   |
|  | #16 AWG Shielded Twisted Pair Cable                                                 |             |     |      | -          | not applicable      |
|  | 20mmØ IMC Pipe                                                                      | Electronics | set | 392  | 140.00     | 54,880.00           |
|  | 25mmØ IMC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 50mmØ IMC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 20mmØ PVC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 25mmØ PVC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 50mmØ PVC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | Pullbox 250mm(H) x 250mm(W) x 150mm(D)                                              | Electronics | set | 42   | 3,250.00   | 136,500.00          |
|  | Junction Box                                                                        | Electronics | set |      |            | -                   |
|  | Others (Please specify in details)                                                  | Electronics | set |      | -          | -                   |
|  | Programming                                                                         | Electronics | set | 2    | 200,000.00 | 400,000.00          |
|  | Supports and Consumables                                                            | Electronics | set | 2    | 41,000.00  | 82,000.00           |
|  |                                                                                     |             |     |      | -          |                     |
|  | <b>Sub-Total Section : Access Control and Monitoring System (Roughing-ins only)</b> |             |     |      | -          | <b>1,821,627.50</b> |
|  |                                                                                     |             |     |      | -          |                     |
|  | <b>CCTV System (Rough-ins Only)</b>                                                 |             |     |      | -          |                     |
|  | To supply and install the complete CCTV System.:                                    |             |     |      | -          |                     |
|  | CCTV Workstation                                                                    | Electronics | set |      | -          |                     |
|  | CCTV Rack                                                                           | Electronics | set |      | -          |                     |
|  | Network Video Recorder                                                              | Electronics | set |      | -          |                     |
|  | Server                                                                              | Electronics | set |      | -          |                     |
|  | CCTV Video Display Monitor                                                          | Electronics | set |      | -          |                     |
|  | Power Supply with UPS                                                               | Electronics | set |      | -          |                     |
|  | POE Switch                                                                          | Electronics | set |      | -          |                     |
|  | Network Switch                                                                      | Electronics | set |      | -          |                     |
|  | CCTV Camera (Dome Type)                                                             | Electronics | set |      | -          |                     |
|  | CCTV Camera (Bullet Type)                                                           | Electronics | set |      | -          |                     |
|  | #24 AWG CAT6 Shielded Twisted Pair Cable                                            | Electronics | set |      | -          |                     |
|  | Others (Please specify in details)                                                  | Electronics | set |      | -          |                     |
|  |                                                                                     |             |     |      | -          |                     |
|  | Supply and Installation of Conduit Fittings and Supports                            |             |     |      | -          |                     |
|  | 20mmØ IMC Pipe                                                                      | Electronics | set | 5460 | 125.00     | 682,500.00          |
|  | 25mmØ IMC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 50mmØ IMC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 20mmØ PVC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 25mmØ PVC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | 50mmØ PVC Pipe                                                                      | Electronics | set |      |            | -                   |
|  | Pullbox 250mm(H) x 250mm(W) x 150mm(D)                                              | Electronics | set |      |            | -                   |
|  | Junction Box                                                                        | Electronics | set |      |            | -                   |
|  | Others (Please specify in details)                                                  | Electronics | set |      | -          | -                   |
|  | Supports and Consumables                                                            | Electronics | set | 2    | 110,000.00 | 220,000.00          |
|  |                                                                                     |             |     |      | -          | -                   |
|  | <b>Sub-Total Section : CCTV System (Roughing-ins only)</b>                          |             |     |      | -          | <b>902,500.00</b>   |
|  |                                                                                     |             |     |      | -          |                     |
|  | <b>Water Leak Detection System</b>                                                  |             |     |      | -          |                     |
|  | To supply and install the complete Water Leak Detection System.:                    |             |     |      | -          |                     |
|  | Water Leak Detection Panel                                                          | Electronics | set | 2    | 25,000.00  | 50,000.00           |
|  | Sensing Cable                                                                       | Electronics | set | 4    | 40,000.00  | 160,000.00          |
|  | Non-Sensing Cable                                                                   | Electronics | set | 2    | 4,050.00   | 8,100.00            |
|  | End-of-Line Resistor                                                                | Electronics | set | 4    | 2,535.00   | 10,140.00           |
|  | Cable Clip/Holder                                                                   | Electronics | set | 12   | 2,535.00   | 30,420.00           |
|  | #24 AWG CAT6 Shielded Twisted Pair Cable                                            | Electronics | set |      | -          | -                   |
|  | Others (Please specify in details)                                                  | Electronics | set |      | -          | -                   |

|  |                                                                                                         |             |      |      |            |                     |
|--|---------------------------------------------------------------------------------------------------------|-------------|------|------|------------|---------------------|
|  | Audio Visual Alarm                                                                                      | Electronics | set  | 2    | 10,140.00  | 20,280.00           |
|  | Modular Branch Connector                                                                                | Electronics | set  | 4    | 8,100.00   | 32,400.00           |
|  | Buzzer & Battery Cabinet (w/o battery)                                                                  | Electronics | set  | 2    | 12,600.00  | 25,200.00           |
|  | Identification Tags for sensor cable, bag of 50                                                         | Electronics | set  | 2    | 3,800.00   | 7,600.00            |
|  | Mapping Tool for TT1000 and TT3000 sensing cable                                                        | Electronics | set  | 2    | 5,070.00   | 10,140.00           |
|  |                                                                                                         |             |      |      | -          | -                   |
|  | Supply and Installation of Conduit Fittings and Supports                                                |             |      |      | -          | -                   |
|  | 20mmØ IMC Pipe                                                                                          | Electronics | set  | 2    | 125.00     | 250.00              |
|  | 25mmØ IMC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 50mmØ IMC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 20mmØ PVC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 25mmØ PVC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 50mmØ PVC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | Pullbox 250mm(H) x 250mm(W) x 150mm(D)                                                                  | Electronics | set  |      |            | -                   |
|  | Junction Box                                                                                            | Electronics | set  |      |            | -                   |
|  | Others (Please specify in details)                                                                      | Electronics | set  |      |            | -                   |
|  | Consumables                                                                                             |             | lot  | 2    | 42,250.00  | 84,500.00           |
|  | Frieght                                                                                                 |             | lot  | 2    | 25,000.00  | 50,000.00           |
|  |                                                                                                         |             |      |      | -          | -                   |
|  | <b>Sub-Total Section : Water Leak Detection System</b>                                                  |             |      |      | -          | <b>489,030.00</b>   |
|  |                                                                                                         |             |      |      | -          |                     |
|  | <b>Building Management System</b>                                                                       |             |      |      | -          |                     |
|  | To supply and install the complete Building Management System and interface with Fire Detection System. |             |      |      | -          |                     |
|  | BMS Worstation                                                                                          | Electronics | set  |      | -          |                     |
|  | BMS Software                                                                                            | Electronics | set  |      | -          |                     |
|  | BMS Panel                                                                                               | Electronics | lot  |      | -          |                     |
|  | BMS Controller                                                                                          | Electronics | set  |      | -          |                     |
|  | Digital Controller                                                                                      | Electronics | set  |      | -          |                     |
|  | Gateway/Translator                                                                                      | Electronics | set  |      | -          |                     |
|  | Switch/Hub                                                                                              | Electronics | set  |      | -          |                     |
|  | Field Devices                                                                                           | Electronics | set  |      | -          |                     |
|  | Temperature Sensor                                                                                      | Electronics | set  |      | -          |                     |
|  | Humidity Sensor                                                                                         | Electronics | set  |      | -          |                     |
|  | Relay/Contactor                                                                                         | Electronics | set  |      | -          |                     |
|  | Others (kindly specify)                                                                                 | Electronics | set  |      | -          |                     |
|  | #16 2/C Shielded Twisted Pair Wires                                                                     | Electronics | lm   | 1200 | 120.00     | 144,000.00          |
|  | #18 2/C Shielded Twisted Pair Wires                                                                     | Electronics | lm   |      |            | -                   |
|  | 20mmØ IMC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 25mmØ IMC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 50mmØ IMC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | Pullbox 250mm(H) x 250mm(W) x 150mm(D)                                                                  | Electronics | set  |      |            | -                   |
|  | Junction Box                                                                                            | Electronics | set  |      |            | -                   |
|  | Devices including all necessary boxes & dropped FM                                                      | Electronics | set  |      |            | -                   |
|  | Miscellaneous/Consumables                                                                               | Electronics | lot  |      |            | -                   |
|  | Testing and Commisssioning                                                                              | Electronics | lot  |      |            | -                   |
|  | Other Accessories for the complete operation of the s                                                   | Electronics |      |      |            | -                   |
|  | Others (Please specify in details)                                                                      | Electronics |      |      |            | -                   |
|  | 200 x 50 cable trunking                                                                                 | Electronics | m    |      |            | -                   |
|  | As-built drawings                                                                                       | Electronics | lot  |      |            |                     |
|  | Operation and Maintenance Manual                                                                        | Electronics | lot  |      |            |                     |
|  | Testing and Commissioning                                                                               | Electronics | lot  |      |            |                     |
|  | CO2 Monitor                                                                                             |             | sets | 68   | 26,000.00  | 1,768,000.00        |
|  | Devs, Panels, Controllers                                                                               |             | lot  | 2    | 400,000.00 | 800,000.00          |
|  |                                                                                                         |             |      |      | -          |                     |
|  | <b>Sub-Total Section : Building Management System</b>                                                   |             |      |      | -          | <b>2,712,000.00</b> |
|  |                                                                                                         |             |      |      | -          |                     |
|  | <b>Audio-Visual System (Rough-ins Only)</b>                                                             |             |      |      | -          |                     |
|  | To supply and install the complete CCTV System.:                                                        |             |      |      | -          |                     |
|  | HDMI Cable                                                                                              | Electronics | set  |      | -          |                     |
|  | AV Encoder/Decoder                                                                                      | Electronics | set  |      | -          |                     |
|  | #24 AWG CAT6 Shielded Twisted Pair Cable                                                                | Electronics | set  |      | -          |                     |
|  | Others (Please specify in details)                                                                      | Electronics | set  |      | -          |                     |
|  |                                                                                                         |             |      |      | -          |                     |
|  | Supply and Installation of Conduit Fittings and Supports                                                |             |      |      | -          |                     |
|  | 20mmØ IMC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 25mmØ IMC Pipe                                                                                          | Electronics | set  | 120  | 195.00     | 23,400.00           |
|  | 50mmØ IMC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 20mmØ PVC Pipe                                                                                          | Electronics | set  |      |            | -                   |
|  | 25mmØ PVC Pipe                                                                                          | Electronics | set  |      |            | -                   |



|              |                                                                                                                                                                           |             |     |      |           |                      |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----|------|-----------|----------------------|
|              | 50mmØ PVC Pipe                                                                                                                                                            | Electronics | set |      |           | -                    |
|              | Pullbox 250mm(H) x 250mm(W) x 150mm(D)                                                                                                                                    | Electronics | set |      |           | -                    |
|              | Junction Box                                                                                                                                                              | Electronics | set | 20   | 210.00    | 4,200.00             |
|              | Others (Please specify in details)                                                                                                                                        |             |     |      | -         | -                    |
|              | Consumables, Supports and Miscellaneous                                                                                                                                   | Electronics | set | 2    | 16,900.00 | 33,800.00            |
|              |                                                                                                                                                                           |             |     |      | -         | -                    |
|              |                                                                                                                                                                           |             |     |      | -         | -                    |
|              | <b>Sub-Total Section : Audio-Visual System (Roughing-ins only)</b>                                                                                                        |             |     |      | -         | <b>61,400.00</b>     |
|              |                                                                                                                                                                           |             |     |      | -         |                      |
|              | <b>Miscellaneous</b>                                                                                                                                                      |             |     |      | -         |                      |
|              | To supply and install all necessary supports/brackets, boxes, etc.                                                                                                        |             |     |      | -         |                      |
|              | Supports, Brackets etc..                                                                                                                                                  | Electronics | lot | 2    | 70,000.00 | 140,000.00           |
|              | Junction Box                                                                                                                                                              | Electronics | lot | 2    | 42,250.00 | 84,500.00            |
|              | Utility Box                                                                                                                                                               | Electronics | lot | 2    | 16,900.00 | 33,800.00            |
|              | Pull Box                                                                                                                                                                  | Electronics | lot | 2    | 16,900.00 | 33,800.00            |
|              | Others (Please specify in details)                                                                                                                                        | Electronics | lot |      | -         | -                    |
|              |                                                                                                                                                                           |             |     |      |           |                      |
|              | <b>Sub-Total Section : Miscellaneous</b>                                                                                                                                  |             |     |      |           | <b>292,100.00</b>    |
|              |                                                                                                                                                                           |             |     |      |           |                      |
|              | <b>Grand Total Electronics System Package</b>                                                                                                                             |             |     |      |           | <b>17,286,523.50</b> |
|              |                                                                                                                                                                           |             |     |      |           |                      |
| <b>17.00</b> | <b>Automatic Fire Sprinkler System</b>                                                                                                                                    |             |     |      |           |                      |
|              | <b>Basic Materials and Methods</b>                                                                                                                                        |             |     |      |           |                      |
|              | <b>Existing Installation</b>                                                                                                                                              |             |     |      |           |                      |
|              | Relocation of existing sprinkler heads and accessories including pipework, supports and other required accessories.                                                       | FP          | Lot | 2    | 27,000.00 | 54,000.00            |
|              | Removal/deletion of existing sprinkler heads and accessories including capping-off of existing branch pipe, charging and restoration of the entire system into operation. | FP          | Lot | 2    | 18,000.00 | 36,000.00            |
|              | Installation/relocation of portable fire extinguishers including accessories required                                                                                     | FP          | Lot | 2    | 4,500.00  | 9,000.00             |
|              |                                                                                                                                                                           |             |     |      | -         |                      |
|              |                                                                                                                                                                           |             |     |      | -         |                      |
|              | <b>New Installation</b>                                                                                                                                                   |             |     |      | -         |                      |
|              | Furnish, install, testing and commission of new sprinkler heads and accessories including pipework and connection to existing sprinkler pipes.                            | FP          | Lot | 2    | -         | included             |
|              | Preparation of As-built Drawings, Test Reports, Operational/Maintenance Manuals and Documents. Provide necessary training to client representatives.                      | FP          | Lot | 2    | -         | included             |
|              |                                                                                                                                                                           |             |     |      | -         |                      |
|              | a. Pendent Sprinkler heads                                                                                                                                                | FP          | pc  | 514  | 775.00    | 398,350.00           |
|              | b. Upright Sprinkler heads                                                                                                                                                | FP          | pc  | 104  | 775.00    | 80,600.00            |
|              |                                                                                                                                                                           |             |     |      | -         | -                    |
|              | Black-iron pipe, Sch. 40 ASTM A53                                                                                                                                         |             |     |      | -         | -                    |
|              | a. Ø25                                                                                                                                                                    | FP          | lm  | 2336 | 180.00    | 420,480.00           |
|              | b. Ø32                                                                                                                                                                    | FP          | lm  | 284  | 273.00    | 77,532.00            |
|              | c. Ø40                                                                                                                                                                    | FP          | lm  | 84   | 360.00    | 30,240.00            |
|              | c. Ø50                                                                                                                                                                    | FP          | lm  | 96   | 360.00    | 34,560.00            |
|              |                                                                                                                                                                           |             |     |      | -         | -                    |
|              | Fittings                                                                                                                                                                  |             |     |      | -         | -                    |
|              | Mechanical Tee Reducer                                                                                                                                                    |             |     |      | -         | -                    |
|              | a. 100Ø x 25Ø                                                                                                                                                             | FP          | pc  | 8    | 1,092.00  | 8,736.00             |
|              | b. 100Ø x 32Ø                                                                                                                                                             | FP          | pc  | 12   | 1,092.00  | 13,104.00            |
|              | c. 100Ø x 40Ø                                                                                                                                                             | FP          | pc  | 18   | 1,092.00  | 19,656.00            |
|              | c. 100Ø x 50Ø                                                                                                                                                             | FP          | pc  | 6    | 1,092.00  | 6,552.00             |
|              |                                                                                                                                                                           |             |     |      | -         | -                    |
|              | Elbow Standard                                                                                                                                                            |             |     |      | -         | -                    |
|              | a. 25Ø                                                                                                                                                                    | FP          | pc  | 1216 | 65.00     | 79,040.00            |
|              | b. 32Ø                                                                                                                                                                    | FP          | pc  | 16   | 100.00    | 1,600.00             |
|              |                                                                                                                                                                           |             |     |      | -         | -                    |
|              | Tee Standard                                                                                                                                                              |             |     |      | -         | -                    |
|              | a. 25Ø                                                                                                                                                                    | FP          | pc  | 128  | 85.00     | 10,880.00            |
|              | b. 32Ø                                                                                                                                                                    | FP          | pc  | 8    | 140.00    | 1,120.00             |
|              | c. 40Ø                                                                                                                                                                    | FP          | pc  | 14   | 172.00    | 2,408.00             |
|              | d. 50Ø                                                                                                                                                                    | FP          | pc  | 8    | 255.00    | 2,040.00             |
|              |                                                                                                                                                                           |             |     |      | -         | -                    |
|              | Tee Reducer                                                                                                                                                               |             |     |      | -         | -                    |
|              | a. 40Ø x 32Ø                                                                                                                                                              | FP          | pc  | 8    | 225.00    | 1,800.00             |
|              | b. 32Ø x 25Ø                                                                                                                                                              | FP          | pc  | 148  | 155.00    | 22,940.00            |



|              |                                                                                                                                                                                             |      |      |        |              |                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|--------------|---------------------|
|              | c. 40Ø x 25Ø                                                                                                                                                                                | FP   | pc   | 24     | 209.00       | 5,016.00            |
|              |                                                                                                                                                                                             |      |      |        | -            | -                   |
|              | Coupling Reducer                                                                                                                                                                            |      |      |        | -            | -                   |
|              | a. 40Ø x 32Ø                                                                                                                                                                                | FP   | pc   | 24     | 227.00       | 5,448.00            |
|              | b. 32Ø x 25Ø                                                                                                                                                                                | FP   | pc   | 96     | 140.00       | 13,440.00           |
|              | c. 25Ø x 15Ø                                                                                                                                                                                | FP   | pc   | 870    | 227.00       | 197,490.00          |
|              | d. 40Ø x 25Ø                                                                                                                                                                                |      |      |        | -            | -                   |
|              | e. 50Ø x 40Ø                                                                                                                                                                                | FP   | pc   | 8      | 227.00       | 1,816.00            |
|              |                                                                                                                                                                                             |      |      |        | -            | -                   |
|              | Portable Fire Extinguishers                                                                                                                                                                 |      |      |        | -            | -                   |
|              | k. 10 lbs (4.5 kg) FE-36 (UL/FM Approved)                                                                                                                                                   | FP   | Nos. | 240    | 9,750.00     | 2,340,000.00        |
|              | l. 10 lbs CO2 Portable Fire Extinguisher                                                                                                                                                    | FP   | Nos. | 10     | 3,900.00     | 39,000.00           |
|              |                                                                                                                                                                                             |      |      |        | -            | -                   |
|              | <b>Clean Agent Fire Suppression System (Novec 1230)</b>                                                                                                                                     |      |      |        | -            | -                   |
|              | A. Clean-Agent Gas Suppression System (including equipment, required fitting, valves, accessories and distribution pipings)                                                                 | FP   | pc   | 2      | 1,350,000.00 | 2,700,000.00        |
|              | Inclusive of the ff.:                                                                                                                                                                       |      |      |        | -            |                     |
|              | a. Smoke Detectors                                                                                                                                                                          | FP   | lb   | 2      | -            | included            |
|              | b. 360 deg. Discharge Nozzle                                                                                                                                                                | FP   | pc   | 2      | -            | included            |
|              | c. Manual Release Station                                                                                                                                                                   | FP   | pc   | 2      | -            | included            |
|              | d. Abort Switch                                                                                                                                                                             | FP   | pc   | 2      | -            | included            |
|              | e. Alarm Bell                                                                                                                                                                               | FP   | pc   | 2      | -            | included            |
|              | f. Horn Strobe                                                                                                                                                                              | FP   | pc   | 2      | -            | included            |
|              | g. Clean-Agent Control Panel                                                                                                                                                                | FP   | pc   | 2      | -            | included            |
|              | h. Gas Fired Illuminate Sign                                                                                                                                                                | FP   | pc   | 2      | -            | included            |
|              | i. Evacuate Area Illuminate Sign                                                                                                                                                            | FP   | pc   | 2      | -            | included            |
|              | j. Pipes and Fittings                                                                                                                                                                       | FP   | pc   | 2      | -            | included            |
|              | k. Electrical conduits and wires                                                                                                                                                            | FP   | pc   | 2      | -            | included            |
|              | l. Testing and Commissioning                                                                                                                                                                | FP   | pc   | 2      | -            | included            |
|              | m. 130 lbs. Capacity Cylinder Tank filled with 111 lbs.                                                                                                                                     | FP   | pc   | 2      | -            | included            |
|              | n. 130 lbs. Capacity Cylinder Tank filled with 94 lbs. N                                                                                                                                    | FP   | pc   | 2      | -            | included            |
|              | o. Novec 1230 Agent                                                                                                                                                                         | FP   | pc   | 2      | -            | included            |
|              | p. Backup Battery, Rechargeable                                                                                                                                                             | FP   | pc   | 2      | -            | included            |
|              | q. Pressure Switch                                                                                                                                                                          | FP   | lot  | 2      | -            | included            |
|              |                                                                                                                                                                                             |      |      |        | -            |                     |
|              | <b>Other Items Required (Please Specify)</b>                                                                                                                                                |      |      |        | -            |                     |
|              | c. Ø65 Sch 40 BI Pipe                                                                                                                                                                       | FP   | lm   |        |              | -                   |
|              | c. Ø80 Sch 40 BI Pipe                                                                                                                                                                       | FP   | lm   |        |              | -                   |
|              | c. Ø100 Sch 40 BI Pipe                                                                                                                                                                      | FP   | lm   |        |              | -                   |
|              | c. Ø50x32 Coupling Reducer                                                                                                                                                                  |      | pcs  | 40     | 280.00       | 11,200.00           |
|              | c. Ø50x25 Coupling Reducer                                                                                                                                                                  |      | pcs  | 122    | 280.00       | 34,160.00           |
|              | Fire Hose Cabinet                                                                                                                                                                           |      | set  | 16     | 14,500.00    | 232,000.00          |
|              | Sidewall Sprinkler                                                                                                                                                                          |      | pcs  | 16     | 773.00       | 12,368.00           |
|              | Supports (clevis, Rod, Dynabolt)                                                                                                                                                            |      | lot  | 2      | 78,500.00    | 157,000.00          |
|              | Fire Rated Flex Hose                                                                                                                                                                        |      | set  | 514    | 1,365.00     | 701,610.00          |
|              | Miscellaneous and consumables                                                                                                                                                               |      | lot  | 2      | 113,000.00   | 226,000.00          |
|              |                                                                                                                                                                                             |      |      |        |              |                     |
|              |                                                                                                                                                                                             |      |      |        |              |                     |
|              | <b>Total Fire Protection System Package</b>                                                                                                                                                 |      |      |        |              | <b>7,987,186.00</b> |
|              |                                                                                                                                                                                             |      |      |        |              |                     |
| <b>18.00</b> | <b>Mechanical Trade Package</b>                                                                                                                                                             |      |      |        |              |                     |
|              | <b>Basic Materials and Methods</b>                                                                                                                                                          |      |      |        |              |                     |
|              | <b>Air Distribution System</b> (Supply and installation of ductwork system including ducts, air plenums, volume dampers, motorised volume dampers, hangers, supports and other accessories) |      |      |        |              |                     |
|              | Polyeliffin insulation 25mmm thick fire retardant type (AC supply, AC return , Exhaust and Fresh Air ducts)                                                                                 | Mech | m2   | 1852.4 | 720.00       | 1,333,728.00        |
|              | MDF/IDF/UPS Underslab insulation 50mm thick rubber insulation fire retardant Class "O"                                                                                                      | Mech | m2   | 60     | 2,700.00     | 162,000.00          |
|              | Motorized smoke and fire damper Combination (MFSD)                                                                                                                                          |      |      |        | -            | -                   |
|              | 1200 x 500                                                                                                                                                                                  | Mech | Set  | 2      | 15,650.00    | 31,300.00           |
|              | 900 x 400                                                                                                                                                                                   | Mech | Set  | 2      | 13,300.00    | 26,600.00           |
|              | 200 x 150                                                                                                                                                                                   |      |      | 4      | 10,615.00    | 42,460.00           |
|              | Supply Air grille, c/w VD Double Deflection type                                                                                                                                            |      |      |        | -            | -                   |
|              | Supply Air grille/diffuser/Linear, c/w VD                                                                                                                                                   |      |      |        | -            | -                   |
|              | 500 x 500 mm neck size                                                                                                                                                                      | Mech | pcs  | 2      | 2,700.00     | 5,400.00            |
|              | 400 x 400 mm neck size                                                                                                                                                                      | Mech | pcs  | 2      | 1,830.00     | 3,660.00            |
|              | 300 x 300 mm neck size                                                                                                                                                                      | Mech | pcs  | 32     | 1,135.00     | 36,320.00           |
|              | 200 x 200 mm neck size                                                                                                                                                                      | Mech | pcs  | 2      | 785.00       | 1,570.00            |
|              | 1200 x 2 SLOTS mm neck size                                                                                                                                                                 | Mech | pcs  | 2      | 1,150.00     | 2,300.00            |
|              | Return Air grille/diffuser/Linear, c/w VD                                                                                                                                                   |      |      |        | 1,150.00     | -                   |
|              | 500 x 500 mm neck size                                                                                                                                                                      | Mech | pcs  | 2      | 2,730.00     | 5,460.00            |

|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       |              |                      |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|-------|--------------|----------------------|
|  | 400 x 400 mm neck size                                                                                                                                                                                                                                                                                                             | Mech        | pcs        | 2     | 1,840.00     | 3,680.00             |
|  | 200 x 200 mm neck size                                                                                                                                                                                                                                                                                                             | Mech        | pcs        | 2     | 780.00       | 1,560.00             |
|  | 1200 x 2 SLOTS mm neck size                                                                                                                                                                                                                                                                                                        | Mech        | pcs        | 2     | 1,150.00     | 2,300.00             |
|  | Exhaust Air grille c/w VD                                                                                                                                                                                                                                                                                                          |             |            | 0     |              | -                    |
|  | 300 x 300 mm neck size                                                                                                                                                                                                                                                                                                             | Mech        | pcs        | 40    | 1,030.00     | 41,200.00            |
|  | 200 x 200 mm neck size                                                                                                                                                                                                                                                                                                             | Mech        | pcs        | 8     | 877.00       | 7,016.00             |
|  | Volume damper                                                                                                                                                                                                                                                                                                                      |             |            |       | -            | -                    |
|  | 250x200 mm                                                                                                                                                                                                                                                                                                                         |             |            |       | -            | -                    |
|  | 200x150 mm                                                                                                                                                                                                                                                                                                                         | Mech        | pcs        | 2     | 395.00       | 790.00               |
|  | 200x200 mm                                                                                                                                                                                                                                                                                                                         | Mech        | pcs        | 14    | 415.00       | 5,810.00             |
|  | 150x150 mm                                                                                                                                                                                                                                                                                                                         | Mech        | pcs        | 6     | 295.00       | 1,770.00             |
|  | Back draft Dampers Two(2) hour fire rated, U.L., F.M Approved                                                                                                                                                                                                                                                                      |             |            |       | -            | -                    |
|  | 200 x 200 mm                                                                                                                                                                                                                                                                                                                       | Mech        | pcs        | 20    | 414.00       | 8,280.00             |
|  | 200 x 150 mm                                                                                                                                                                                                                                                                                                                       | Mech        | pcs        | 26    | 395.00       | 10,270.00            |
|  | 150 x 150 mm                                                                                                                                                                                                                                                                                                                       | Mech        | pcs        | 116   | 297.00       | 34,452.00            |
|  | Others to complete ( Contractor to itemized)                                                                                                                                                                                                                                                                                       |             |            |       | -            | -                    |
|  | MFSD, 150X150                                                                                                                                                                                                                                                                                                                      | Mech        | pcs        | 4     | 10,615.00    | 42,460.00            |
|  | Volume Damper, 1100x400                                                                                                                                                                                                                                                                                                            | Mech        | pcs        | 8     | 10,770.00    | 86,160.00            |
|  | Exhaust Air Louver, 200x150                                                                                                                                                                                                                                                                                                        | Mech        | pcs        | 2     | 850.00       | 1,700.00             |
|  | Fire Damper, 150x150                                                                                                                                                                                                                                                                                                               | Mech        | pcs        | 2     | 8,300.00     | 16,600.00            |
|  | TDC Ducts (EAD and SAD)                                                                                                                                                                                                                                                                                                            |             |            |       | -            | -                    |
|  | GA#26                                                                                                                                                                                                                                                                                                                              | Mech        | sqm        | 1219  | 840.00       | 1,023,960.00         |
|  | GA#24                                                                                                                                                                                                                                                                                                                              | Mech        | sqm        | 627.9 | 930.00       | 583,947.00           |
|  | GA#22                                                                                                                                                                                                                                                                                                                              | Mech        | sqm        | 662.4 | 1,120.00     | 741,888.00           |
|  | GA#20                                                                                                                                                                                                                                                                                                                              | Mech        | sqm        | 243.8 | 1,050.00     | 255,990.00           |
|  | GA#18                                                                                                                                                                                                                                                                                                                              | Mech        | sqm        | 595.7 | 1,430.00     | 851,851.00           |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | <b>SUB-TOTAL Section :Air Distribution System</b>                                                                                                                                                                                                                                                                                  |             |            |       |              | <b>5,372,482.00</b>  |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | <b>VRF Piping system</b> (Supply and installation of Copper pipes, HD type L, including rubber insulation, vapor barrier minimum 3/4 in thick Fire rated , supports and miscellaneous materials and other accessories- Contractor to coordinate with VRF equipment manufacturer for refrigrant copper pipe sizes prior to bidding) |             |            |       | -            |                      |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | 6.35 mm diameter                                                                                                                                                                                                                                                                                                                   | Mech        | lm         |       |              | -                    |
|  | 9.52 mm diameter                                                                                                                                                                                                                                                                                                                   | Mech        | lm         |       |              | -                    |
|  | 12.70 mm diameter                                                                                                                                                                                                                                                                                                                  | Mech        | lm         |       |              | -                    |
|  | 15.88 mm diameter                                                                                                                                                                                                                                                                                                                  | Mech        | lm         |       |              | -                    |
|  | 19.05 mm diameter                                                                                                                                                                                                                                                                                                                  | Mech        | lm         |       |              | -                    |
|  | 22.20 mm diameter                                                                                                                                                                                                                                                                                                                  | Mech        | lm         |       |              | -                    |
|  | 28.58 mm diameter                                                                                                                                                                                                                                                                                                                  | Mech        | lm         |       |              | -                    |
|  | 34.93 mm diameter                                                                                                                                                                                                                                                                                                                  | Mech        | lm         |       |              | -                    |
|  | 41.28 mm diameter                                                                                                                                                                                                                                                                                                                  | Mech        | lm         |       |              | -                    |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       |              |                      |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | Condensate drains system complete with rubber insulation minimum 1/2 inch thick vapor barrier,                                                                                                                                                                                                                                     | Mech        | lot        | 2     | 130,000.00   | 260,000.00           |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | Labor and materials for hauling from Ground level to designated areas of Contactor <b>supplied VRF FCUs and ACCUs including installation to its permanent position</b> complete with Equipment                                                                                                                                     | Mech        | lot        | 2     | 95,000.00    | 190,000.00           |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | Labor and materials for the installation of VRF system central monitoring and control system including control wirings conduits roughing ins homerun to new facility office at <b>4th</b> floor.                                                                                                                                   | Mech        | lot        | 2     | 1,200,000.00 | 2,400,000.00         |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | ATC 1000 Auto Change over at Data Center                                                                                                                                                                                                                                                                                           |             |            |       | -            |                      |
|  | ATC 1000 Auto Change over at AVR Room                                                                                                                                                                                                                                                                                              |             |            |       | -            |                      |
|  | VRF Central Controller (to incude existng landlord VRF)                                                                                                                                                                                                                                                                            | Mech        | lot        |       | -            |                      |
|  | Others (Please specify in details)                                                                                                                                                                                                                                                                                                 |             |            |       | -            |                      |
|  | Refrigerant Piping, Insulation, Suports and installation of FCU, CU Units                                                                                                                                                                                                                                                          | Mech        | lot        | 2     | 4,950,000.00 | 9,900,000.00         |
|  | RefNet, Drain Pump Kit, Piping Kit, Remote controls, Standard Panel, Decoration Panel                                                                                                                                                                                                                                              | Mech        | lot        | 2     | 1,350,000.00 | 2,700,000.00         |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | <b>SUB-TOTAL Section : VRV Piping Distribution System</b>                                                                                                                                                                                                                                                                          | <b>Mech</b> | <b>lot</b> |       | <b>-</b>     | <b>15,450,000.00</b> |
|  |                                                                                                                                                                                                                                                                                                                                    |             |            |       | -            |                      |
|  | <b>Mechanical Equipment</b>                                                                                                                                                                                                                                                                                                        |             |            |       | -            |                      |

|  |                                                                                                                                                                                                                                                                                                    |      |     |   |           |           |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|---|-----------|-----------|
|  | <b>Contractor Supplied ACMV Equipment</b><br>(Mechanical Contractor to quote for works needed to incorporate the equipment to system. Mechanical Contractor to coordinate and assist vendors during installation, testing and commissioning)                                                       |      |     |   | -         |           |
|  | FCU-4F-01-01A VRF,4-WAY Ceiling cassette type, 7.1kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 25,340.00 | 50,680.00 |
|  | FCU-4F-01-01B VRF,4-WAY Ceiling cassette type, 7.1kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 25,340.00 | 50,680.00 |
|  | FCU-4F-01-02 VRF, ceiling concealed ducted type, 2.2kw cooling cap., 0.04kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 23,500.00 | 47,000.00 |
|  | FCU-4F-01-03 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 24,500.00 | 49,000.00 |
|  | FCU-4F-01-04 VRF,4-WAY Ceiling cassette type, 3.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 24,500.00 | 49,000.00 |
|  | FCU-4F-01-05 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 18,000.00 | 36,000.00 |
|  | FCU-4F-01-06 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 18,000.00 | 36,000.00 |
|  | FCU-4F-01-07 VRF,4-WAY Ceiling cassette type, 3.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 22,500.00 | 45,000.00 |
|  | FCU-4F-01-08 VRF, ceiling concealed ducted type, 5.6kw cooling cap., 0.055kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 26,000.00 | 52,000.00 |

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|  | FCU-4F-01-09 VRF , ceiling concealed ducted type, 5.6kw cooling cap., 0.055kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 26,000.00 | 52,000.00 |
|  | FCU-4F-01-10A VRF,4-WAY Ceiling cassette type, 9kw cooling cap., 0.05kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.       | Mech | set | 2 | 27,500.00 | 55,000.00 |
|  | FCU-4F-01-10B VRF,4-WAY Ceiling cassette type, 9kw cooling cap., 0.05kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.       | Mech | set | 2 | 27,500.00 | 55,000.00 |
|  | FCU-4F-01-10C VRF,4-WAY Ceiling cassette type, 9kw cooling cap., 0.05kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.       | Mech | set | 2 | 27,500.00 | 55,000.00 |
|  | FCU-4F-01-10D VRF,4-WAY Ceiling cassette type, 9kw cooling cap., 0.05kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.       | Mech | set | 2 | 27,500.00 | 55,000.00 |
|  | FCU-4F-01-10E VRF,4-WAY Ceiling cassette type, 9kw cooling cap., 0.05kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.       | Mech | set | 2 | 27,500.00 | 55,000.00 |
|  | FCU-4F-01-10F VRF,4-WAY Ceiling cassette type, 9kw cooling cap., 0.05kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.       | Mech | set | 2 | 27,500.00 | 55,000.00 |
|  | FCU-4F-01-11 VRF,4-WAY Ceiling cassette type, 3.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.      | Mech | set | 2 | 22,500.00 | 45,000.00 |
|  | FCU-4F-01-12 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 18,000.00 | 36,000.00 |
|  | FCU-4F-01-13 VRF,4-WAY Ceiling cassette type, 5.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.      | Mech | set | 2 | 24,700.00 | 49,400.00 |

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|  | FCU-4F-01-14 VRF,4-WAY Ceiling cassette type, 4.5kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.      | Mech | set | 2 | 24,000.00 | 48,000.00 |
|  | FCU-4F-01-15 VRF , ceiling concealed ducted type, 11.2kw cooling cap., 0.16kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 24,000.00 | 48,000.00 |
|  | FCU-4F-02-01 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-02-02 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-02-03 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-02-04 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-02-05 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-02-06 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-02-07 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-02-08 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibaration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 28,500.00 | 57,000.00 |

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|  | FCU-4F-02-09 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-03-01 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-03-02 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-03-03 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-03-04 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-03-05 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-03-06 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-03-07 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00 |
|  | FCU-4F-03-08 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00 |
|  | FCU-4F-03-09 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 23,373.00 | 46,746.00 |

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| FCU-4F-03-10 VRF, 4-WAY Ceiling cassette type, 4.5kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 24,000.00 | 48,000.00  |
| FCU-4F-03-11 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00  |
| FCU-4F-03-12 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00  |
| FCU-4F-03-13 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00  |
| FCU-4F-03-14 VRF, ceiling concealed ducted type, 16kw cooling cap., 0.19kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 56,977.20 | 113,954.40 |
| FCU-4F-03-15 VRF, 4-WAY Ceiling cassette type, 3.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 22,500.00 | 45,000.00  |
| FCU-4F-03-16 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00  |
| FCU-4F-03-17 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00  |
| FCU-4F-04-01 VRF, 4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00  |
| FCU-4F-04-02 VRF, 4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00  |



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|  | FCU-4F-04-03 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-04 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-05 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-06 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-07 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-08 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-09 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-10 VRF,4-WAY Ceiling cassette type, 11.2kw cooling cap., 0.07kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,500.00 | 57,000.00 |
|  | FCU-4F-04-11 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00 |
|  | FCU-4F-04-12 VRF,1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 18,000.00 | 36,000.00 |

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| FCU-4F-04-13 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 18,000.00 | 36,000.00 |
| FCU-4F-05-01 VRF, 4-WAY Ceiling cassette type, 3.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 22,500.00 | 45,000.00 |
| FCU-4F-05-02 VRF, 4-WAY Ceiling cassette type, 3.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 22,500.00 | 45,000.00 |
| FCU-4F-05-03 VRF, 4-WAY Ceiling cassette type, 4.5kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 24,000.00 | 48,000.00 |
| FCU-4F-05-04 VRF, 4-WAY Ceiling cassette type, 5.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 24,500.00 | 49,000.00 |
| FCU-4F-05-05 VRF, 4-WAY Ceiling cassette type, 4.5kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 24,000.00 | 48,000.00 |
| FCU-4F-05-06 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 18,000.00 | 36,000.00 |
| FCU-4F-05-07 VRF, 1-WAY Ceiling cassette type, 2.2kw cooling cap., 0.042kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 18,000.00 | 36,000.00 |
| FCU-4F-05-08 VRF, 4-WAY Ceiling cassette type, 14kw cooling cap., 0.11kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 29,000.00 | 58,000.00 |
| FCU-4F-05-09 VRF, ceiling concealed ducted type, 11.2kw cooling cap., 0.16kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 28,000.00 | 56,000.00 |

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| FCU-4F-05-10A VRF,4-WAY Ceiling cassette type, 5.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 24,500.00  | 49,000.00  |
| FCU-4F-05-10B VRF,4-WAY Ceiling cassette type, 5.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 24,500.00  | 49,000.00  |
| FCU-4F-05-10C VRF,4-WAY Ceiling cassette type, 5.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 24,500.00  | 49,000.00  |
| FCU-4F-05-11 VRF,4-WAY Ceiling cassette type, 3.6kw cooling cap., 0.03kw/230v/1ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.    | Mech | set | 2 | 24,500.00  | 49,000.00  |
| PFCU-4F-06-01 VRF , ceiling concealed ducted type, 28kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 120,000.00 | 240,000.00 |
| PFCU-4F-06-02 VRF , ceiling concealed ducted type, 28kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 120,000.00 | 240,000.00 |
| PFCU-4F-06-03 VRF , ceiling concealed ducted type, 28kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 120,000.00 | 240,000.00 |
| PFCU-4F-07-01 VRF , ceiling concealed ducted type, 28kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 0 | -          | -          |
| PFCU-4F-07-01 VRF , ceiling concealed ducted type, 28kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 120,000.00 | 240,000.00 |
| PFCU-4F-07-02 VRF , ceiling concealed ducted type, 28kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 120,000.00 | 240,000.00 |

|                                                                                                                                                                                                                                                                                                      |      |     |   |              |              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|---|--------------|--------------|
| PFCU-4F-07-03 VRF, ceiling concealed ducted type, 28kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.     | Mech | set | 2 | 120,000.00   | 240,000.00   |
| FCU-4F-HUB-01 VRF, ceiling concealed ducted type, 22.4kw cooling cap., 0.63kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 4 | 34,000.00    | 136,000.00   |
| FCU-4F-HUB-02 VRF, ceiling concealed ducted type, 22.4kw cooling cap., 0.63kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 34,000.00    | 68,000.00    |
| FCU-4F-AVR-01A VRF, ceiling concealed ducted type, 22.4kw cooling cap., 0.63kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 34,000.00    | 68,000.00    |
| FCU-4F-AVR-01B VRF, ceiling concealed ducted type, 22.4kw cooling cap., 0.63kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 34,000.00    | 68,000.00    |
| FCU-4F-AVR-02 VRF, ceiling concealed ducted type, 22.4kw cooling cap., 0.63kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 34,000.00    | 68,000.00    |
| ACCU-4F-01, Floor Mounted, 124kw cooling cap., 32.37kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.                         | Mech | set | 2 | 1,350,000.00 | 2,700,000.00 |
| ACCU-4F-02, Floor Mounted, 101.5kw cooling cap., 24.57kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.                       | Mech | set | 2 | 490,000.00   | 980,000.00   |
| ACCU-4F-03, Floor Mounted, 113.5kw cooling cap., 28.8kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.                        | Mech | set | 2 | 300,000.00   | 600,000.00   |
| ACCU-4F-04, Floor Mounted, 120kw cooling cap., 29.62kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.                         | Mech | set | 2 | 1,400,000.00 | 2,800,000.00 |
| ACCU-4F-05, Floor Mounted, 73.5kw cooling cap., 18.32kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.                        | Mech | set | 2 | 615,000.00   | 1,230,000.00 |

|  |                                                                                                                                                                                                                                                                                 |      |     |   |            |              |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|---|------------|--------------|
|  | ACCU-4F-06, Floor Mounted, 90.4kw cooling cap., 29.62kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 540,000.00 | 1,080,000.00 |
|  | ACCU-4F-07, Floor Mounted, 90.4kw cooling cap., 29.62kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.   | Mech | set | 2 | 615,000.00 | 1,230,000.00 |
|  | ACCU-4F-HUB-01, Floor Mounted, 28kw cooling cap., 7.08kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 34,000.00  | 68,000.00    |
|  | ACCU-4F-HUB-02, Floor Mounted, 28kw cooling cap., 7.08kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 34,000.00  | 68,000.00    |
|  | ACCU-4F-AVR-01, Floor Mounted, 55kw cooling cap., 18.39kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set | 2 | 34,000.00  | 68,000.00    |
|  | ACCU-4F-AVR-02, Floor Mounted, 28kw cooling cap., 7.08kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set | 2 | 34,000.00  | 68,000.00    |
|  | EF-4F-01, 530 l/s, in-line centrifugal, 150Pa ESP, 0.144kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                            | Mech | set | 2 | 39,000.00  | 78,000.00    |
|  | EF-4F-02, 150 l/s, in-line centrifugal, 150Pa ESP, 0.04kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                             | Mech | set | 2 | 5,500.00   | 11,000.00    |
|  | EF-4F-03, 105 l/s, in-line centrifugal, 150Pa ESP, 0.04kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                             | Mech | set | 2 | 5,500.00   | 11,000.00    |
|  | EF-4F-04, 40 l/s, Ceiling Cassette type, 150Pa ESP, 0.024kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                           | Mech | set | 2 | 3,900.00   | 7,800.00     |
|  | EF-4F-05, 40 l/s, in-line centrifugal, 150Pa ESP, 0.024kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                             | Mech | set | 2 | 3,900.00   | 7,800.00     |
|  | EF-4F-06, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                           | Mech | set | 2 | 1,550.00   | 3,100.00     |
|  | EF-4F-07, 25 l/s, Ceiling Cassette type, 150Pa ESP, 0.025kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                           | Mech | set | 2 | 1,550.00   | 3,100.00     |
|  | EF-4F-08, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                           | Mech | set | 2 | 1,550.00   | 3,100.00     |
|  | EF-4F-09, 70 l/s, in-line centrifugal, 150Pa ESP, 0.04kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                              | Mech | set | 2 | 1,500.00   | 3,000.00     |
|  | EF-4F-10, 30 l/s, Ceiling Cassette type, 150Pa ESP, 0.025kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                           | Mech | set | 2 | 3,900.00   | 7,800.00     |
|  | EF-4F-11, 30 l/s, Ceiling Cassette type, 150Pa ESP, 0.025kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                           | Mech | set | 2 | 1,550.00   | 3,100.00     |
|  | EF-4F-12, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)                                                                                                                                           | Mech | set | 2 | 1,550.00   | 3,100.00     |

|  |                                                                                                                                                |      |     |   |            |            |
|--|------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|---|------------|------------|
|  | EF-4F-13, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)          | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | EF-4F-14, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)          | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | EF-4F-15, 130 l/s, Ceiling Cassette type, 150Pa ESP, 0.061kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)         | Mech | set | 2 | 5,900.00   | 11,800.00  |
|  | EF-4F-16, 130 l/s, Ceiling Cassette type, 150Pa ESP, 0.061kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)         | Mech | set | 2 | 5,900.00   | 11,800.00  |
|  | EF-4F-17, 50 l/s, Ceiling Cassette type, 150Pa ESP, 0.024kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)          | Mech | set | 2 | 3,900.00   | 7,800.00   |
|  | EF-4F-18, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)          | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | EF-4F-19, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)          | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | EF-4F-20, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)          | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | EF-4F-21, 20 l/s, Ceiling Cassette type, 150Pa ESP, 0.013kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)          | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | TEF-4F-01, 25 l/s, Ceiling Cassette type, 150Pa ESP, 0.025kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)         | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | TEF-4F-02, 25 l/s, Ceiling Cassette type, 150Pa ESP, 0.025kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)         | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | TEF-4F-03, 25 l/s, Ceiling Cassette type, 150Pa ESP, 0.025kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)         | Mech | set | 2 | 1,550.00   | 3,100.00   |
|  | GEF-4F-01, 105 l/s, in-line centrifugal type, 150Pa ESP, 0.04kw/230v/1ph/60hz (Complete with vibration isolators, motor Control, Starter)      | Mech | set | 2 | 5,400.00   | 10,800.00  |
|  | SEF-5F-01, 11600 l/s, SISW CABINET FAN BELT DRIVEN, 500Pa ESP, 7.5kw/380v/3ph/60hz (Complete with vibration isolators, motor Control, Starter) | Mech | set | 2 | 300,000.00 | 600,000.00 |
|  | AP-4F-01 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-02 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-03 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-04 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-05 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-06 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-07 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-08 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-09 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |
|  | AP-4F-10 176 l/s, Floor mounted(movable), 0.14kw/230v/1ph/60hz (Complete with HEPA Fiilter and accessories required by supplier)               | Mech | set | 2 | 40,000.00  | 80,000.00  |

[illegible]

|  |                                                                                                                                                                                                                                                                                                   |      |      |    |            |                      |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|----|------------|----------------------|
|  | FCU-4F-08-01 VRF, ceiling concealed ducted type, 3.6kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set  | 2  | 40,000.00  | 80,000.00            |
|  | FCU-4F-08-02 VRF, ceiling concealed ducted type, 11.2kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set  | 2  | 45,000.00  | 90,000.00            |
|  | FCU-4F-08-03 VRF, ceiling concealed ducted type, 22.4kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment. | Mech | set  | 2  | 55,000.00  | 110,000.00           |
|  | FCU-4F-08-04 VRF, ceiling concealed ducted type, 3.6kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set  | 2  | 40,000.00  | 80,000.00            |
|  | PFCU-4F-08-01 VRF, ceiling concealed ducted type, 14kw cooling cap., 0.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.  | Mech | set  | 2  | 130,000.00 | 260,000.00           |
|  | ACCU-4F-08, Floor Mounted, 28kw cooling cap., 73.5kw/380v/3ph/60hz. Complete with piping insulation , room thermostat, control and power wirings, supports, vibration isolators miscellaneous materials including start up testing and commissioning of the A/C equipment.                        | Mech | set  | 2  | 80,000.00  | 160,000.00           |
|  | <b>SUB-TOTAL Section : Quote for Contractor Supplied ACMV Equipment and Installation</b>                                                                                                                                                                                                          |      |      |    | -          | <b>22,551,460.40</b> |
|  |                                                                                                                                                                                                                                                                                                   |      |      |    | -          | -                    |
|  | <b>Systems Testing commissioning and Balancing</b>                                                                                                                                                                                                                                                |      |      |    | -          | -                    |
|  | Complete Testing and balancing of the VRF A/C and Ventilation system                                                                                                                                                                                                                              | Mech | lot  | 2  | 155,000.00 | 310,000.00           |
|  | Others to complete (Contractor to Itemized)                                                                                                                                                                                                                                                       | Mech | pcs  |    | -          | -                    |
|  | Daiken FCU / CU                                                                                                                                                                                                                                                                                   |      | lot  | 2  | 155,000.00 | 310,000.00           |
|  |                                                                                                                                                                                                                                                                                                   |      |      |    | -          | -                    |
|  | <b>SUB-TOTAL Section :Systems Testing commissioning &amp; Balancing</b>                                                                                                                                                                                                                           |      |      |    | -          | <b>620,000.00</b>    |
|  |                                                                                                                                                                                                                                                                                                   |      |      |    | -          | -                    |
|  | <b>Others to Complete MVAC System</b> (Please note: Base Building Contractor will provide Pipe/ duct chase, demolition works for duct risers , Demolition for Refrigerant pipe risers and punch-thru-wall or horizontal run will be by this Contractor)                                           |      |      |    | -          | -                    |
|  | Grouting of openings in floors and walls, after pipes in place & sealing of such openings when not used, (L & M) in place.                                                                                                                                                                        | Mech | lot  | 2  | 15,500.00  | 31,000.00            |
|  | Hacking of non-structural walls,floors and partitions for pipes opening between sleeves and pipes.                                                                                                                                                                                                | Mech | lot  | 2  | 1,900.00   | 3,800.00             |
|  | Restoration /repair of existing structures damaged during installation.                                                                                                                                                                                                                           | Mech | lot  | 2  | 9,300.00   | 18,600.00            |
|  | Miscellaneous items and other provision required to maintain cleanliness level of adjacent areas during the entire duration of construction. (See general requirements)                                                                                                                           | Mech | lot  | 2  | 35,000.00  | 70,000.00            |
|  | ACCU and EF Concrete Pads                                                                                                                                                                                                                                                                         | Mech | sets | 26 | 19,500.00  | 507,000.00           |
|  | Others to Complete (Contractor to Itemized)                                                                                                                                                                                                                                                       |      |      |    | -          |                      |



|              |                                                                                                                                                                                            |          |     |     |           |                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-----|-----------|----------------------|
|              | Refrigerant Net Fittings                                                                                                                                                                   |          |     |     | -         |                      |
|              |                                                                                                                                                                                            |          |     |     | -         |                      |
|              |                                                                                                                                                                                            |          |     |     | -         |                      |
|              | <b>SUB_TOTAL Section : Others for MVAC</b>                                                                                                                                                 |          |     |     | -         | <b>630,400.00</b>    |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | <b>TOTAL SECTION 2.2: MECHANICAL TRADE PACKAGES</b>                                                                                                                                        |          |     |     |           | <b>44,624,342.40</b> |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
| <b>19.00</b> | <b>Sanitary/Plumbing System</b>                                                                                                                                                            |          |     |     |           |                      |
|              | <b>Basic Materials and Methods</b>                                                                                                                                                         |          |     |     |           |                      |
|              | <b>Water Supply System</b>                                                                                                                                                                 |          |     |     |           |                      |
|              | Supply and installation of polypropylene (PPR) pipe, PN20, including sleeves, hangers and supports, painting and other miscellaneous items as shown and as required to complete the system |          |     |     |           |                      |
|              | 40 mm diameter                                                                                                                                                                             | Plumbing | lm  | 144 | 1,105.00  | 159,120.00           |
|              | 32 mm diameter                                                                                                                                                                             | Plumbing | lm  |     |           | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | lm  |     |           | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | lm  | 48  | 475.00    | 22,800.00            |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | lm  | 132 | 475.00    | 62,700.00            |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of PPR elbow 90°                                                                                                                                                   |          |     |     | -         | -                    |
|              | 40 mm diameter                                                                                                                                                                             | Plumbing | pcs |     | -         | -                    |
|              | 32 mm diameter                                                                                                                                                                             | Plumbing | pcs |     | -         | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs |     | -         | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs |     | -         | -                    |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of PPR coupling                                                                                                                                                    |          |     |     | -         | -                    |
|              | 40 mm diameter                                                                                                                                                                             | Plumbing | pcs | 36  | 240.00    | 8,640.00             |
|              | 32 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs | 12  | 142.00    | 1,704.00             |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | pcs | 32  | 142.00    | 4,544.00             |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of PPR Tee                                                                                                                                                         |          |     |     | -         | -                    |
|              | 40 mm diameter                                                                                                                                                                             | Plumbing | pcs | 2   | 237.00    | 474.00               |
|              | 32 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | pcs | 2   | 142.00    | 284.00               |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of PPR Male Threaded Adaptor                                                                                                                                       |          |     |     | -         | -                    |
|              | 40 mm diameter                                                                                                                                                                             | Plumbing | pcs |     | -         | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs |     | -         | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs |     | -         | -                    |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | pcs | 2   | 142.00    | 284.00               |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of PPR Female Threaded Adaptor                                                                                                                                     |          |     |     | -         | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of PPR Female Threaded Elbow Adaptor                                                                                                                               |          |     |     |           | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of Gate Valve                                                                                                                                                      |          |     |     | -         | -                    |
|              | 40 mm diameter                                                                                                                                                                             | Plumbing | pcs | 4   | 2,055.00  | 8,220.00             |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs |     |           | -                    |
|              | 15 mm diameter                                                                                                                                                                             | Plumbing | pcs | 8   | 711.00    | 5,688.00             |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of Water Hammer Arrester                                                                                                                                           |          |     |     | -         | -                    |
|              | 25 mm diameter                                                                                                                                                                             | Plumbing | pcs | 8   | 1,105.00  | 8,840.00             |
|              | 20 mm diameter                                                                                                                                                                             | Plumbing | pcs | 8   | 1,105.00  | 8,840.00             |
|              |                                                                                                                                                                                            |          |     |     |           |                      |
|              | Supply and installation of Electric Water Heater                                                                                                                                           |          |     |     | -         | -                    |
|              | 6 KW                                                                                                                                                                                       | Plumbing | pcs | 4   | 26,900.00 | 107,600.00           |
|              |                                                                                                                                                                                            |          |     |     | -         |                      |
|              | <b>SUB-TOTAL Section : Water Supply System</b>                                                                                                                                             |          |     |     | -         | <b>399,738.00</b>    |
|              |                                                                                                                                                                                            |          |     |     | -         |                      |
|              | <b>Sanitary, Waste &amp; Vent System</b>                                                                                                                                                   |          |     |     | -         |                      |
|              | Supply and installation of polyvinyl chloride (PVC) Pipes, series 1000, including sleeves, painting, supports, hangers, and other miscellaneous items as shown                             |          |     |     |           |                      |
|              |                                                                                                                                                                                            |          |     |     | -         |                      |

|                                    |                                                              |                 |            |          |                   |                       |
|------------------------------------|--------------------------------------------------------------|-----------------|------------|----------|-------------------|-----------------------|
|                                    | and as required to complete the system.                      |                 |            |          | -                 |                       |
|                                    |                                                              |                 |            |          | -                 |                       |
|                                    | 100 mm diameter                                              | Plumbing        | lm         | 240      | 632.00            | 151,680.00            |
|                                    | 75 mm diameter                                               | Plumbing        | lm         | 240      | 474.00            | 113,760.00            |
|                                    | 50 mm diameter                                               | Plumbing        | lm         | 240      | 395.00            | 94,800.00             |
|                                    |                                                              |                 |            |          | -                 | -                     |
|                                    | Supply and installation PVC of elbow 90°                     |                 |            |          | -                 | -                     |
|                                    | 100 mm diameter                                              | Plumbing        | pcs        | 20       | 285.00            | 5,700.00              |
|                                    | 75 mm diameter                                               | Plumbing        | pcs        | 12       | 285.00            | 3,420.00              |
|                                    | 50 mm diameter                                               | Plumbing        | pcs        | 12       | 285.00            | 3,420.00              |
|                                    |                                                              |                 |            |          | -                 | -                     |
|                                    | Supply and installation of PVC elbow 45°                     |                 |            |          | -                 | -                     |
|                                    | 100 mm diameter                                              | Plumbing        | pcs        | 20       | 285.00            | 5,700.00              |
|                                    | 75 mm diameter                                               | Plumbing        | pcs        | 12       | 285.00            | 3,420.00              |
|                                    | 50 mm diameter                                               | Plumbing        | pcs        | 12       | 285.00            | 3,420.00              |
|                                    |                                                              |                 |            |          | -                 | -                     |
|                                    | Supply and installation of PVC Wye                           |                 |            |          | -                 | -                     |
|                                    | 100 mm diameter                                              | Plumbing        | pcs        | 20       | 505.00            | 10,100.00             |
|                                    | 75 mm diameter                                               | Plumbing        | pcs        | 12       | 347.00            | 4,164.00              |
|                                    | 50 mm diameter                                               | Plumbing        | pcs        | 12       | 142.00            | 1,704.00              |
|                                    |                                                              |                 |            |          | -                 | -                     |
|                                    | Supply and installation of PVC Wye Reducer                   |                 |            |          | -                 | -                     |
|                                    | 100 x 50 mm diameter                                         | Plumbing        | pcs        | 8        | 505.00            | 4,040.00              |
|                                    | 75 x 50 mm diameter                                          | Plumbing        | pcs        | 8        | 347.00            | 2,776.00              |
|                                    |                                                              |                 |            |          | -                 | -                     |
|                                    | Supply and installation of TEE Standard                      |                 |            |          | -                 | -                     |
|                                    | 50 mm diameter                                               | Plumbing        | pcs        |          |                   | -                     |
|                                    |                                                              |                 |            |          |                   | -                     |
|                                    | Supply and installation of TEE Reducer                       |                 |            |          |                   | -                     |
|                                    | 100 x 50 mm diameter                                         | Plumbing        | pcs        |          |                   | -                     |
|                                    | 75 x 50 mm diameter                                          | Plumbing        | pcs        |          |                   | -                     |
|                                    |                                                              |                 |            |          |                   | -                     |
|                                    | Supply and installation of P-Trap                            |                 |            |          |                   | -                     |
|                                    | 50 mm diameter                                               | Plumbing        | pcs        |          |                   | -                     |
|                                    |                                                              |                 |            |          |                   | -                     |
|                                    | Supply and installation of Floor Drain                       |                 |            |          |                   | -                     |
|                                    | 50 mm diameter                                               | Plumbing        | pcs        |          |                   | -                     |
|                                    |                                                              |                 |            |          |                   | -                     |
|                                    | Supply and installation of Funnel Drain                      |                 |            |          |                   | -                     |
|                                    | 50 mm diameter                                               | Plumbing        | pcs        |          |                   | -                     |
|                                    |                                                              |                 |            |          |                   | -                     |
|                                    | Supply and installation of Floor Clean-out                   |                 |            |          |                   | -                     |
|                                    | 100 mm diameter                                              | Plumbing        | pcs        |          |                   | -                     |
|                                    | 75 mm diameter                                               | Plumbing        | pcs        |          |                   | -                     |
|                                    | 50 mm diameter                                               |                 |            |          |                   | -                     |
|                                    |                                                              |                 |            |          |                   | -                     |
|                                    | Supply and installation of Grease Trap                       |                 |            |          | -                 | -                     |
|                                    | 15 GPM                                                       | Plumbing        | pcs        | 4        | 47,400.00         | 189,600.00            |
|                                    |                                                              |                 |            |          | -                 |                       |
|                                    |                                                              |                 |            |          | -                 |                       |
|                                    | <b>SUB-TOTAL Section : Sanitary, Waste &amp; Vent System</b> |                 |            |          | -                 | <b>597,704.00</b>     |
|                                    |                                                              |                 |            |          | -                 |                       |
|                                    |                                                              |                 |            |          | -                 |                       |
|                                    | <b>Other Items Required (Please Specify)</b>                 | Plumbing        | set        |          | -                 | -                     |
|                                    | <b>Miscellaneous Consumables, Coring, etc</b>                | <b>Plumbing</b> | <b>lot</b> | <b>2</b> | <b>130,000.00</b> | <b>260,000.00</b>     |
|                                    |                                                              |                 |            |          | -                 | -                     |
|                                    | <b>Grand Total Plumbing System Package</b>                   | <b>Plumbing</b> |            |          |                   | <b>1,257,442.00</b>   |
|                                    |                                                              |                 |            |          |                   |                       |
|                                    |                                                              |                 |            |          |                   |                       |
| <b>GRAND TOTAL (VAT Inclusive)</b> |                                                              |                 |            |          |                   | <b>104,926,237.90</b> |

Prepared by:

Engr. Mark Ralphael P. Tive  
MRK - QS

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